

Generation Interconnection System Impact Study Report

For Queue Project Q21-1744

North Findlay 138 kV

July 16, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1744. The Interconnection Customer proposes a 127.0 MW natural gas combined cycle facility interconnecting at the North Findlay 138 kV substation in the Kentucky area. The request was received on May 11, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 568 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,403,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
FLEWELLEN to BAKER 138kV	BIG SANDY PEAKE PLANT to CEREDO GENERATING STA 138kV	118.3	402	100.0
N FINDLAY to FLEWELLEN 138kV	BIG SANDY PEAKER PLANT to CEREDO GENERATING STA 138kV L	117.6	412	100.0
KRS II KOLOA SOLAR to CEREDO GENERATING STATION 138kV	CEREDO GENERATING STATION to TRISTATE 138kV	103.7	341	62.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
FLEWELLEN to BAKER 138kV	Rebuild line with higher-capacity conductor	\$15,092,000	25% (\$3,714,000)
N FINDLAY to FLEWELLEN 138kV	Rebuild line with higher-capacity conductor	\$9,938,000	25% (\$2,446,000)
CEREDO GENERATING STATION to KRS II KOLOA SOLAR 138kV	Reconductor line	\$1,135,000	21% (\$243,000)
	Total allocated to project		\$6,403,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.